Graph Neural Network-Based Link Scheduling for Dense Uncrewed Aerial Vehicle Swarms

Qingzhe Zeng, Lin Bai, Rui Han, Jiaxing Wang, Xin Xie, and Guowei Shi

***Abstract*—Dense low-altitude uncrewed aerial vehicle (UAV) networks require efficient link scheduling to coordinate concurrent transmissions under strong and highly asymmetric mutual interference. The core of maximizing single-hop medium access control (MAC) throughput lies in balancing *spatial reuse* (activating multiple sufficiently separated links) with *multiuser diversity* (opportunistically selecting links with the best instantaneous channel states). This paper studies two related scheduling problems to systematically navigate this trade-off from physical-layer capacity limits to MAC-layer execution. We first formulate the instantaneous network sum-rate maximization under continuous physical interference. To this end, we develop a graph attention network (GAT)-based scheduler that explores the interference-limited rate envelope, thereby implicitly extracting multi-hop interference representations through a fixed number of local message-passing rounds. Moving beyond the full-buffer assumption, we then introduce heterogeneous traffic arrivals and per-link queues, formulating a cross-layer scheduling problem aimed at maximum stable throughput. We propose a Marginal Utility Message Passing (MarginalMP) protocol that reuses the GAT-derived expected interference representation to construct an initial collision-aware skeleton. This skeleton is subsequently refined through exact queue-weighted marginal coordination. The evaluation framework incorporates persistent link identities, correlated Rician block fading, mobility, imperfect channel state information (CSI), and separate accounting of algorithmic signaling.**

***Index Terms*—Graph attention networks, link scheduling, low-altitude networks, multiuser diversity, queue-aware scheduling, spatial reuse, UAV swarms.**

I. Introduction

DENSE low-altitude UAV networks play a pivotal role in establishing flexible flying ad-hoc networks (FANETs). Unlike traditional terrestrial cellular architectures, FANETs operate in highly dynamic 3D topologies characterized by prevalent line-of-sight (LoS) propagation. As swarm density increases, the massive connectivity inevitably leads to severe, asymmetric mutual interference. Consequently, uncoordinated contention-based medium access control (MAC) protocols, such as carrier-sense multiple access with collision avoidance (CSMA/CA), suffer from exacerbated hidden and exposed terminal problems. Explicit link scheduling thus becomes imperative to determine an optimal active subset of links at each time slot, mitigating mutual interference while maximizing spectrum utilization.

The fundamental challenge in designing such scheduling mechanisms lies in navigating the inherent trade-off between *spatial reuse* and *multiuser diversity*. On the one hand, spatial reuse encourages the simultaneous activation of multiple links separated by sufficient distances, expanding the spatial degrees of freedom. On the other hand, multiuser diversity dictates that amidst severe local contention, the scheduler should selectively activate links experiencing the best instantaneous signal-to-noise ratio (SNR) with minimal external interference footprint, while silencing those in deep fading.

Traditionally, this problem is abstracted as a maximum weighted independent set (MWIS) problem on a binary conflict graph. However, binary protocol models tend to over-penalize weak interference, yielding overly conservative schedules. While exploring the optimal trade-off under continuous signal-to-interference-plus-noise ratio (SINR) constraints provides a critical physical-layer (PHY) sum-rate benchmark, such formulations typically rely on a full-buffer assumption. In practical UAV swarms, traffic arrives stochastically. Activating a high-capacity link with an empty queue wastes valuable scheduling opportunities. To bridge the PHY and MAC layers, the objective must shift toward maximizing long-term delivered throughput subject to queue stability. Driven by classical Maximum Weight Scheduling (MWS) theories, this requires prioritizing congested links on a slot-by-slot basis. Yet, finding an exact distributed MWS solver under non-convex continuous SINR constraints incurs prohibitive signaling overhead that cannot track fast-varying UAV channels.

Recently, graph neural networks (GNNs) have emerged as a compelling paradigm for wireless scheduling. Their inherent permutation equivariance and localized message-passing mechanisms allow them to infer global interference topologies from low-dimensional local observations. To fully exploit this potential without violating strict interaction budgets, we propose a two-stage, GAT-assisted cross-layer scheduling framework. First, under a PHY-oriented full-buffer assumption, a multi-head GAT learns to balance spatial reuse and multiuser diversity, extracting a latent multi-hop interference representation. Second, we transition to practical MAC execution by introducing a fixed-round Marginal Utility Message Passing (MarginalMP) protocol. This step reuses the GAT's output to form a sparse initial skeleton, which is then dynamically fine-tuned via exact queue-weighted marginal coordination, effectively stabilizing network queues while strictly controlling signaling overhead.

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II. Related Work

A. UAV and Low-Altitude Wireless Networks

Prior UAV networking studies have examined measured air-to-air (A2A) propagation, multi-UAV access, trajectory and association design, and edge-service deployment [1]–[6]. Related work on random access and physical-layer transmission also provides useful designs for mobile edge and distributed wireless systems [7]–[9].

B. Wireless Link Scheduling

Classical wireless-network studies established capacity and interference abstractions for multihop networks [10], [11]. Queue-based scheduling, carrier-sense multiple access (CSMA), local greedy methods, and fractional programming provide important references for distributed link activation under different interference assumptions [12]–[16].

C. Learning-Based Wireless Scheduling

Graph learning has been applied to wireless resource allocation and link scheduling because interference relations admit a natural graph representation [17]–[20].

D. Imperfect CSI and Distributed Implementation

III. System Model and Problem Formulation

We consider a dense low-altitude network with a set of UAVs $\mathcal{U} = \{1, \dots, N\}$. Each UAV is equipped with a single antenna and communicates in half-duplex mode over a shared frequency channel. Time is divided into slots of duration τ. At each slot, each link independently decides whether to activate based on local information. Let $\mathcal{L} = \{1, \dots, L\}$ denote the set of candidate directed links, where link $i \in \mathcal{L}$ corresponds to a fixed transmitter–receiver pair (s_i, r_i) with $s_i, r_i \in \mathcal{U}$ and $s_i \neq r_i$. Activated links transmit data from their queues at their achievable rates.

A. Network Topology and Channel Model

UAV positions follow a horizontal mobility model with specular reflection at the boundary. Position of UAV u in block b is $\mathbf{x}_u[b] \in \mathbb{R}^3$, and the distance between UAVs u and v is $d_{uv}[b] = \|\mathbf{x}_u[b] - \mathbf{x}_v[b]\|_2$.

The air-to-air (A2A) channel follows an LoS-dominant Rician block-fading model [1]: the channel remains constant within a block and evolves independently across blocks. Let $b(t)$ denote the channel block containing slot t. The complex channel from UAV u to UAV v in block b is

$$h_{uv}[b] = \sqrt{\lambda_0}\, d_{uv}[b]^{-\xi/2} \left(\sqrt{\frac{K}{K+1}} e^{-j2\pi d_{uv}[b]/\lambda_c} + \sqrt{\frac{1}{K+1}} g_{uv}[b] \right), \tag{1}$$

where $\lambda_0 = (c/(4\pi f_c))^2$ is the free-space propagation constant, f_c is the carrier frequency, $\lambda_c = c/f_c$ is the carrier wavelength, ξ is the path-loss exponent, K is the linear Rician factor, and $g_{uv}[b] \sim \mathcal{CN}(0,1)$ is the scattering component.

The instantaneous received power at UAV v from UAV u is $P^{\mathrm{rx}}_{uv}[b] = P_{\mathrm{tx}}|h_{uv}[b]|^2$, and its mean over the scattering component is $\overline{P}^{\mathrm{rx}}_{uv}[b] = P_{\mathrm{tx}}\lambda_0 d_{uv}[b]^{-\xi}$.

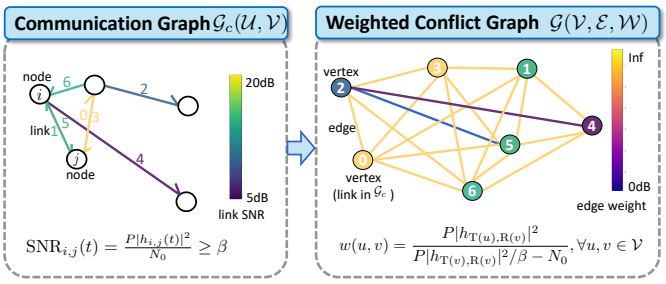


Fig. 1. Mapping from communication graph to directed conflict graph.

The scattering component is constant within a block and evolves across blocks as

$$g_{uv}[b] = \rho\, g_{uv}[b-1] + \sqrt{1-\rho^2}\, \epsilon_{uv}[b], \quad \epsilon_{uv}[b] \sim \mathcal{CN}(0,1), \tag{2}$$

where ρ controls the inter-block correlation.

A link catalog $\mathcal{L}$ is established in the initial channel block based on the mean received power. Link i is detectable in block b when

$$\nu_i[b] = \mathbb{1}\left\{ \frac{\overline{P}^{\mathrm{rx}}_{s_i r_i}[b]}{N_0} \geq \beta \right\}, \tag{3}$$

where N_0 is the receiver noise power spectral density and β is the SNR sensitivity threshold [9]. An undetectable link remains in the catalog but cannot be scheduled.

B. Interference and Scheduling Model

Classical link scheduling models interference as a maximum weighted independent set (MWIS) problem: the candidate links form a binary undirected conflict graph, where adjacent links impose hard exclusion. In a dense network, this binary symmetric abstraction can yield an overly sparse schedule by suppressing links whose mutual interference is actually tolerable. To address this, we adopt a weighted directed interference graph to retain interference direction and strength, and evaluate rates under continuous physical interference.

The candidate links form a weighted directed interference graph $\mathcal{G}[b] = (\mathcal{L}, \mathcal{E}[b], \mathcal{W}[b])$, as illustrated in Fig.1. A directed edge $j \to i$, where $i, j \in \mathcal{L}$, indicates that the transmitter of link j can significantly affect the receiver of link i. Following the physical-interference model in [11], the edge exists if the two links share a UAV or if

$$P^{\mathrm{rx}}_{s_j r_i}[b] \geq \frac{P^{\mathrm{rx}}_{s_i r_i}[b]}{\beta} - N_0. \tag{4}$$

For a physical-interference edge, the weight $w^{\mathrm{raw}}_{ji}[b]$ quantifies the interference strength and is defined as

$$w^{\mathrm{raw}}_{ji}[b] = \frac{P^{\mathrm{rx}}_{s_j r_i}[b]}{P^{\mathrm{rx}}_{s_i r_i}[b]/\beta - N_0}. \tag{5}$$

Links sharing a UAV are assigned a prescribed maximum weight, constituting hard conflicts under the half-duplex constraint. All other edges represent soft physical interference.

At slot t, each link produces a binary activation variable $a_i(t) \in \{0,1\}$, where $a_i(t) = 1$ indicates activation. Let $\mathbf{a}(t) = [a_1(t), \ldots, a_L(t)]^{\mathsf{T}}$. The feasible activation set is

$$\mathcal{A}[b] = \left\{ \mathbf{a} \in \{0,1\}^L : \begin{array}{c} a_i \le \nu_i[b],\ \forall i \in \mathcal{L}, \\ \sum_{i:u\in\{s_i,r_i\}} a_i \le 1,\ \forall u \in \mathcal{U} \end{array} \right\}. \quad (6)$$

The first constraint ensures that a link is detectable in the current block, and the second constraint enforces the half-duplex rule.

For $\mathbf{a}(t) \in \mathcal{A}[b(t)]$, the signal-to-interference-plus-noise ratio (SINR) of active link i is

$$\gamma_i(\mathbf{a}(t); \mathbf{P}[b(t)]) = \frac{P^{\mathrm{rx}}_{s_i r_i}[b(t)]}{N_0 + \sum_{j\neq i} a_j(t) P^{\mathrm{rx}}_{s_j r_i}[b(t)]}. \quad (7)$$

Here $\mathbf{P}[b(t)] \triangleq (P^{\mathrm{rx}}_{uv}[b(t)])_{u,v\in\mathcal{U}}$ collects all instantaneous received powers in block $b(t)$ and serves as the channel-state input to γ_i. The link rate $R_i(t) = R_i(\mathbf{a}(t); \mathbf{P}[b(t)])$ follows from γ_i via Shannon's formula.

C. Traffic and Queueing Model

The instantaneous link rate $R_i(\mathbf{a}(t); \mathbf{P}[b(t)])$ is determined jointly by the scheduling decision $\mathbf{a}(t)$ and the power-vector state $\mathbf{P}[b(t)]$. In practice, activated links transmit packets that have accumulated in the upper-layer queues. Section V introduces stochastic traffic arrivals and per-link queue backlogs, so scheduling must balance transmission capability with backlog demand.

Each link $i \in \mathcal{L}$ maintains a queue with backlog $Q_i(t)$ at the beginning of slot t. Packet arrivals follow an independent Poisson process:

$$A_i(t) \sim \mathrm{Poisson}(\lambda_i), \quad (8)$$

where λ_i may differ across links, yielding heterogeneous traffic load.

Let L_p be the fixed packet size in bits. The service capacity of link i in slot t (in packets) is $S_i(t) = B\tau R_i(\mathbf{a}(t); \mathbf{P}[b(t)])/L_p$. The actual service is $D_i(t) = \min\{Q_i(t), S_i(t)\}$. Service is completed before new arrivals in the same slot, and the queue evolves as

$$Q_i(t+1) = \max\{Q_i(t) - D_i(t), 0\} + A_i(t). \quad (9)$$

D. Problem Formulation

To enhance the overall spectral efficiency of the dense UAV network, we first formulate the instantaneous network sum-rate maximization problem as follows:

$$\begin{aligned} \mathcal{P}_1: \quad & \max_{\mathbf{a}\in\{0,1\}^L} \sum_{i\in\mathcal{L}} R_i(\mathbf{a}; \mathbf{P}[b]), \quad (10) \\ & \text{s.t. } a_i \in \{0,1\}, \quad \forall i \in \mathcal{L}, \\ & \quad a_i \le \nu_i[b], \quad \forall i \in \mathcal{L}, \\ & \quad \sum_{i:u\in\{s_i,r_i\}} a_i \le 1, \quad \forall u \in \mathcal{U}. \end{aligned}$$

The optimum of $\mathcal{P}_1$ explores the performance boundary of spatial reuse from a physical-layer throughput perspective. However, accounting for this optimal active set is computationally demanding. While traditional link scheduling is often abstracted as a Maximum Weighted Independent Set (MWIS) problem, our formulation preserves the precisely quantified interference rather than simplifying it into binary exclusions. This non-convex combinatorial coupling makes it difficult to solve directly using conventional optimization techniques.

While $\mathcal{P}_1$ effectively establishes the limit of spatial multiplexing under a full-buffer assumption, it ignores the actual transmission demands of the upper layers. In practical UAV swarms, traffic demands can be highly heterogeneous across different service areas. If the scheduler is driven solely by PHY achievable rates, links with long-term favorable channel conditions will be scheduled repeatedly. This greedy behavior can severely starve other links that experience heavier traffic pressure but slightly worse channels.

Therefore, the optimization objective can be extended to incorporate Quality of Service (QoS) by considering queue dynamics. Motivated by classical Maximum Weight Scheduling (MWS) and Lyapunov optimization theories [14], adopting a queue-weighted optimization objective can effectively achieve long-term network stability. Accordingly, the scheduling problem is posed as maximizing the long-term expected delivered service over T slots while keeping all per-link queues stable [21]:

$$\begin{aligned} \mathcal{P}_2: \quad & \max_{\{\mathbf{a}(t)\}} \frac{1}{T}\sum_{t=1}^{T}\sum_{i\in\mathcal{L}} \mathbb{E}[D_i(t)], \quad (11) \\ & \text{s.t. } a_i(t) \in \{0,1\}, \quad \forall i \in \mathcal{L}, \\ & \quad a_i(t) \le \nu_i[b(t)], \quad \forall i \in \mathcal{L}, \\ & \quad \sum_{i:u\in\{s_i,r_i\}} a_i(t) \le 1, \quad \forall u \in \mathcal{U}, \\ & \quad \limsup_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T}\sum_{i\in\mathcal{L}} \mathbb{E}[Q_i(t)] < \infty. \end{aligned}$$

By establishing the connection between physical capacity bounds and MAC-layer throughput demands, this two-stage formulation effectively bridges the PHY and MAC layers. We will systematically discuss how to address these two coupled problems via a low-overhead, distributed learning and coordination framework in the subsequent sections.

IV. GAT-Based Link Scheduling for Sum-Rate Maximization

In this section, we formulate the solution for $\mathcal{P}_1$. The inherent graph structure of the wireless interference model naturally inspires the exploration to adopt graph neural networks (GNNs). Given the large-scale, weighted, and directed characteristics of dense conflict graphs, spatial-based GATs are significantly more suitable than traditional spectral-based GCNs, which require prohibitive eigendecomposition. By leveraging the message-passing mechanism, GAT enables the scheduling model to acquire a broad network vision through strictly local information exchanges. This empowers the network to achieve high spatial reuse with remarkably low interaction overhead.

To implement this efficiently, we adopt a centralized training and distributed execution (CTDE) protocol [18], [22]. The

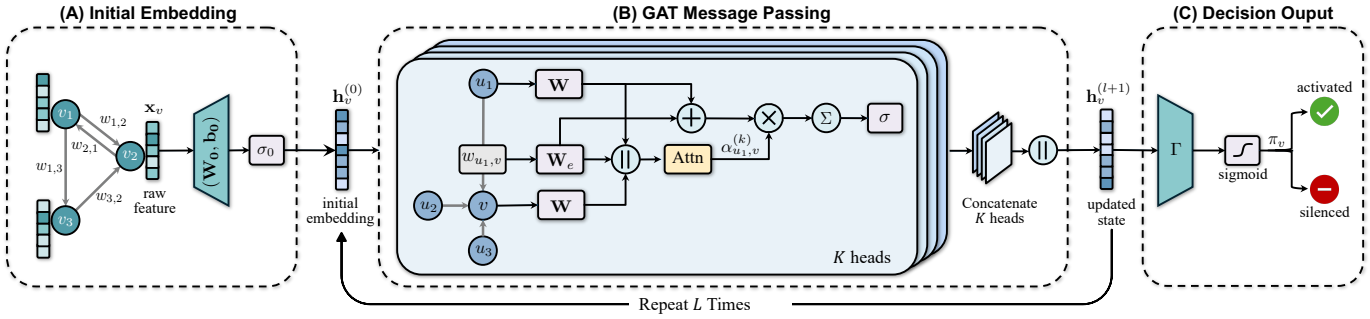


Fig. 2. Illustration of the GAT-based distributed link scheduling framework.

GAT model is trained offline using global network snapshots to capture generalizable interference patterns. Once trained, the model parameters are distributed to the UAVs for online execution. Fig.2 illustrates the overall pipeline of our proposed GAT-assisted scheduling scheme.

A. Graph Representation and Feature Initialization

Each candidate link $i \in \mathcal{L}$ corresponds to a vertex in the conflict graph $\mathcal{G}[b] = (\mathcal{L}, \mathcal{E}[b], \mathcal{W}[b])$. The raw feature of vertex i, denoted as $\mathbf{x}_i[b]$, is the intuitively observable block-constant direct SNR, which is standardized to remove scale variations across networks of different densities. Similarly, for each directed edge $j \to i$, the edge feature $\phi_{ji}[b]$ represents the physical interference, which is also standardized to quantify the extent to which link j degrades the reception of link i.

Following graph embedding architectures [19], an initial multi-layer perceptron (MLP) projects the raw features into a low-dimensional initial vertex embedding:

$$\mathbf{h}_i^{(0)}[b] = \sigma_0\big(\mathbf{W}_0 \mathbf{x}_i[b] + \mathbf{b}_0\big), \tag{12}$$

where $\mathbf{h}_i^{(0)}[b]$ serves as the input to the subsequent GAT message passing, $\sigma_0(\cdot)$ is the activation function, and $\{\mathbf{W}_0, \mathbf{b}_0\}$ are learnable parameters.

B. Edge-Enhanced Graph Attention

Scheduling decisions at every UAV link are produced by an iterative GAT that aggregates information locally over a fixed number of message-passing rounds. Each round expands the effective receptive field beyond the one-hop neighborhood, allowing the vertex to infer complex interference patterns that cannot be observed from its own SNR alone. We denote by $\mathbf{h}_i^{(l)}$ the representation of vertex i at layer l, and let $\mathcal{N}_i$ denote its set of incoming neighbors on the conflict graph.

To properly reflect the asymmetric nature of wireless interference, we modify the standard attention mechanism by explicitly injecting the directed edge feature $\phi_{ji}[b]$. The vertex and edge features are first projected into a shared attention space:

$$\mathbf{q}_i^{(l)} = \mathbf{W}\mathbf{h}_i^{(l)}, \quad \mathbf{k}_j^{(l)} = \mathbf{W}\mathbf{h}_j^{(l)}, \quad \mathbf{e}_{ji}^{(l)} = \mathbf{W}_{\mathrm{e}}\, \phi_{ji}[b].$$

The unnormalized correlation score is evaluated by concatenating these projections:

$$e_{j\to i}^{(l)} = \mathrm{LeakyReLU}\Big(\mathbf{a}^{\mathsf{T}}\big[\mathbf{q}_i^{(l)} \parallel \mathbf{k}_j^{(l)} \parallel \mathbf{e}_{ji}^{(l)}\big]\Big), \tag{13}$$

and the normalized attention coefficient is obtained via a softmax function:

$$\alpha_{j\to i}^{(l)} = \frac{\exp\big(e_{j\to i}^{(l)}\big)}{\sum_{k\in\mathcal{N}_i} \exp\big(e_{k\to i}^{(l)}\big)}. \tag{14}$$

By explicitly incorporating directed edge features, the model ensures that heterogeneous and directional interference relationships are accurately reflected in the neural embeddings.

To capture diverse interference dynamics such as varying geographic distances and rapid small-scale fading, we adopt multi-head attention. By executing K independent attention heads in parallel, the model projects vertex features into multiple distinct representation subspaces. Their outputs are concatenated to form the final update for each iteration:

$$\mathbf{h}_i^{(l+1)}[b] = \Big\|_{k=1}^{K} \sigma\Big(\sum_{j\in\mathcal{N}_i} \alpha_{j\to i}^{(k,l)}\big(\mathbf{W}^{(k)}\mathbf{h}_j^{(l)}[b] + \mathbf{W}_{\mathrm{e}}^{(k)}\,\phi_{ji}[b]\big)\Big). \tag{15}$$

Dense conflict graphs tend to be fully connected, so only a few message-passing iterations are adequate [22]. In practice, $L = 2$ layers suffice to capture dominant local interference without incurring over-smoothing, so that each vertex has an effective receptive field that approximates the global interference structure.

C. Unsupervised Optimization and Final Decision

After L rounds of message passing, the final embedding $\mathbf{h}_i^{(L)}[b]$ is mapped to a continuous activation tendency through an output MLP $\Gamma(\cdot)$ followed by a sigmoid:

$$\tilde{\pi}_i[b] = \sigma\big(\Gamma(\mathbf{h}_i^{(L)}[b])\big) \in (0,1). \tag{16}$$

At deployment, the continuous output is binarized to obtain the final scheduling decision for the vertex:

$$a_i[b] = \mathbb{1}\big\{\tilde{\pi}_i[b] \geq 0.5\big\}. \tag{17}$$

The threshold 0.5 is used throughout this paper.

To preserve differentiability through the rate expression during gradient-based training, the discrete activation variables a_j in the interference term are relaxed to their soft continuous counterparts $\tilde{\pi}_j$. This yields a continuous surrogate rate:

$$\tilde{R}_i[b] = B\log_2\!\left(1 + \frac{\tilde{\pi}_i[b]\, P^{\mathrm{rx}}_{s_i r_i}[b]}{\Gamma\big(N_0 + \sum_{j\in\mathcal{N}_i} \tilde{\pi}_j[b]\, P^{\mathrm{rx}}_{s_j r_i}[b]\big)}\right). \tag{18}$$

Algorithm 1 Distributed GAT-assisted Link Scheduling

Require: Conflict graph $\mathcal{G}[b]$, local node features $\{\mathbf{x}_i[b]\}_{i\in\mathcal{L}}$, layer count L, attention heads K.
Ensure: Discrete link activation decisions $a_i[b] \in \{0, 1\}$.
Initialization Phase:
Extract initial embeddings: $\mathbf{h}_i^{(0)}[b] = \sigma_0(\mathbf{W}_0\mathbf{x}_i[b] + \mathbf{b}_0)$.
Message Passing Phase:
for layer $l = 0$ to $L-1$ **do**
  for each vertex $i \in \mathcal{L}$ **in parallel do**
    for each head $k = 1$ to K **do**
      Compute attention score $e_{j\to i}^{(k,l)}$ and normalized weight $\alpha_{j\to i}^{(k,l)}$ for neighbors $j \in \mathcal{N}_i$ via Eq. (14).
      Aggregate: $\mathbf{z}_i^{(k)} = \sum_{j\in\mathcal{N}_i} \alpha_{j\to i}^{(k,l)} \left(\mathbf{W}^{(k)}\mathbf{h}_j^{(l)}[b] + \mathbf{W}_{\mathrm{e}}^{(k)}\phi_{ji}[b]\right)$.
    end for
    Update state: $\mathbf{h}_i^{(l+1)}[b] = \Big\|_{k=1}^{K} \sigma(\mathbf{z}_i^{(k)})$.
  end for
end for
Decision Output Phase:
for each vertex $i \in \mathcal{L}$ **in parallel do**
  Compute soft probability: $\tilde{\pi}_i[b] = \mathrm{sigmoid}\left(\Gamma(\mathbf{h}_i^{(L)}[b])\right)$.
  Binarize decision: $a_i[b] = \mathbb{1}\{\tilde{\pi}_i[b] \geq 0.5\}$.
end for

The GAT is trained end-to-end against the negative sum-rate of every training snapshot. The unsupervised loss function minimizes the negative expected sum-rate:

$$\min_{\boldsymbol{\theta}} \ \mathcal{L}(\boldsymbol{\theta}) = -\sum_{i\in\mathcal{L}} \tilde{\pi}_i[b]\, \tilde{R}_i[b]. \tag{19}$$

As discussed in our preliminary work [22], stable convergence can be guaranteed under a properly tuned learning rate, eliminating the need for heuristic sparsity or conflict penalties.

D. Distributed Deployment and Complexity

After offline training is completed, the learned parameters are distributed to every UAV. Online inference is fully distributed and operates without any central coordinator. Since the channel features are block-constant under the block-fading model, scheduling decisions are produced locally and only once at the beginning of each channel block. This one-shot execution eliminates the global CSI exchange or central scheduler at runtime, making it highly compatible with the strict latency constraints of fast-varying FANETs. The complete distributed online scheduling procedure is summarized in Algorithm **??**.

To analyze the computational complexity, the scheduling inference involves three stages: initial feature embedding, L rounds of GAT message passing, and the final decision output. The dominant computational overhead lies in the message passing. Summing the per-layer vertex projection Ld^2, per-edge attention-score projection $K|\mathcal{E}|d$, and per-edge aggregation $K|\mathcal{E}|d$ across the L GAT layers yields the total real multiplications per block as:

$$C_{\mathrm{GAT}} = L\big(2Ld^2 + 2K|\mathcal{E}|d\big), \tag{20}$$

TABLE I
COMPUTATIONAL COMPLEXITY COMPARISON (NUMBER OF REAL MULTIPLICATIONS PER SCHEDULING DECISION).

Method	Complexity
Spatial DL [18]	$2L^2K_{\mathrm{conv}}s^2$
FPLinQ [16]	$\frac{1}{2}IL^2(L+1)$
Local Greedy [15]	$L\lvert\mathcal{N}_i\rvert$
GAT scheduler (this paper)	$L\big(2Ld^2 + 2K\lvert\mathcal{E}\rvert d\big)$

where L is the number of candidate links, $|\mathcal{E}|$ the number of directed interference edges, L also the number of GAT layers, K the number of attention heads, and d the embedding dimension per head.

Table I compares the computational complexity of our proposed GAT scheduler against several baselines. Here, I denotes the number of outer WMMSE iterations in FPLinQ; K_{conv} and s represent the number and spatial size of CNN kernels in Spatial DL, respectively. In extremely dense networks where the number of candidate links L is large, our per-link execution cost is bounded by $2K|\mathcal{N}_i|d$ real multiplications per round, which effectively avoids the global $\mathcal{O}(L^2)$ overhead, achieving highly scalable execution while still capturing asymmetric physical interference through the learned attention weights.

V. QUEUE-AWARE LINK SCHEDULING VIA MARGINAL MESSAGE PASSING

The GAT scheduler establishes the optimal spatial-reuse envelope under full-buffer conditions. However, to execute practical PHY-MAC cross-layer scheduling ($\mathcal{P}_2$), the mechanism must dynamically prioritize links with actual data backlogs. According to classical Lyapunov optimization, stabilizing the network queues requires maximizing the queue-weighted delivered service on a slot-by-slot basis. Therefore, we propose the GAT-Assisted MarginalMP framework, which repurposes the GAT's output to construct an initial skeleton and then refines it through exact marginal coordination.

A. GAT-Derived Initial Schedule

To transition from pure capacity to queue-weighted urgency, we first utilize the continuous activation propensities $\tilde{\pi}_j$ to compute a GAT-induced expected-interference score for link i:

$$\widehat{I}_i^{\mathrm{GAT}}(t) = \sum_{j\in\mathcal{N}_i} \tilde{\pi}_j P_{s_j r_i}^{\mathrm{rx}}[b(t)]. \tag{21}$$

This score provides a structural prior of the physical capacity that link i can reliably achieve amidst neighboring contention:

$$C_i^{\mathrm{GAT}}(t) = \log_2\left(1 + \frac{P_{s_i r_i}^{\mathrm{rx}}[b(t)]}{\Gamma(N_0 + \widehat{I}_i^{\mathrm{GAT}}(t))}\right). \tag{22}$$

At slot t, each link combines this capacity prior with its current queue backlog $Q_i(t)$ to calculate its local competitive weight:

$$w_i(t) = Q_i(t)C_i^{\mathrm{GAT}}(t). \tag{23}$$

Through a single local exchange, any link possessing the strictly maximum weight within its conflict neighborhood $\mathcal{N}_i$ joins the initial active set $\mathcal{A}^{(0)}$. This local-max operation guarantees a sparse, collision-aware starting skeleton that immediately favors congested queues while retaining the GAT's spatial reuse insights.

B. Marginal-Utility Message Passing

The initial skeleton $\mathcal{A}^{(0)}$ operates locally and may lack precise multi-hop coordination. To further align the scheduling decisions with the MWS objective without incurring the overhead of exact centralized solvers, we introduce a bounded Marginal Utility Message Passing mechanism.

Let $U(\mathcal{A}) = \sum_{i\in\mathcal{L}} Q_i(t)R_i(\mathcal{A};\mathbf{P}[b(t)])$ denote the global queue-weighted utility for a given active set. In each round k, every link evaluates its exact marginal impact. For an inactive link $i \notin \mathcal{A}^{(k)}$, the marginal joining utility is:

$$\Delta_i^+ = U(\mathcal{A}^{(k)} \cup \{i\}) - U(\mathcal{A}^{(k)}). \tag{24}$$

This term explicitly deducts the physical interference penalty that link i's transmission will inflict on all currently active receivers from the new queue-weighted service link i intends to gain. Conversely, for an active link $i \in \mathcal{A}^{(k)}$, the marginal leaving utility quantifies the network-wide benefit of yielding the channel to relieve interference on others:

$$\Delta_i^- = U(\mathcal{A}^{(k)} \setminus \{i\}) - U(\mathcal{A}^{(k)}). \tag{25}$$

C. Distributed Interface and Complexity

Links with a positive marginal utility ($\Delta_i > 0$) broadcast state-flip proposals. To prevent network oscillations caused by simultaneous conflicting updates, local arbitration retains only the proposal with the highest marginal utility within a localized collision domain. The winning links concurrently flip their states, yielding $\mathcal{A}^{(k+1)}$.

The MarginalMP executes for a fixed budget of $K = 4$ rounds per slot. While motivated by maximum weight scheduling and Lyapunov stability, restricting the interaction rounds ensures bounded control signaling latency, prioritizing practical distributed execution over strict capacity-region optimality. This slot-level negotiation effectively bridges the gap between GAT-derived physical limits and MAC-layer queue realities.

VI. Simulation Results

VII. Conclusion

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